

The Circular Convolution of Prime Constellations in Congruence Phase Space

Ruqing Chen

GUT Geoservice Inc., Montreal, Quebec, Canada

ruqing@hotmail.com

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Abstract

Following the empirical establishment of the m -dimensional proper-depth decay law in Volume I, this paper initiates the theoretical framework of high-dimensional congruence phase-space dynamics. We transition from discrete sieve algorithms to continuous geometric measure theory. By defining the ‘dead(q)’ congruence mechanism as a Castellated Indicator Function on the continuous phase-space torus $\mathbb{T}^1$, we demonstrate that the spatial entanglement of adjacent prime constellations is governed by continuous circular convolution integrals. For Twin-Twin entanglement, this integral evaluates to exactly three quantized interference states. The limit case $j = 1$ (adjacent twins merging into a quadruplet) is pure destructive interference, evaluating to $R(1) = \mathfrak{S}_{\text{quad}}/\mathfrak{S}_{\text{twin}}^2 \approx 0.40$: this quantifies the suppression of the quadruplet *prefactor* (its singular series relative to two independent twins) and is complementary to—not a restatement of—the $-m$ proper-depth slope of Volume I, which is fixed by the member count. The two effects live on orthogonal axes: interference sets the constant, dimension sets the decay rate.

1 Introduction

In the first nineteen parts of this series, the Arithmetic Geodynamics framework relied heavily on high-resolution empirical sieving across decimal shells (up to S_{11}). The core finding was the rigid scaling law: $d \ln \rho_m / d\ell \rightarrow -m$, where $\ell = \ln \ln X$ is the proper depth and m is the tuple dimension [4, 5, 6].

However, to compute the spatial correlation of the $6N$ lattice—specifically the entanglement between a constellation at center N and another at $N + j$ —discrete sieving becomes computationally intractable and theoretically opaque. In this paper, we map the discrete modular arithmetic of the $6N$ skeleton onto a continuous geometric manifold. We show that prime constellation survival can be modeled as the integration of geometric filters over a phase-space torus $\mathbb{T}^1$.

We stress at the outset that this *horizontal* axis (the lag j between centers) is orthogonal to the *vertical* axis of Volume I (the proper depth ℓ). The depth flow $d \ln \rho_m / d\ell \rightarrow -m$ is a statement about the $(\ln X)^{-m}$ scaling, fixed by the number of members; the horizontal correlation studied here is a statement about the X -independent singular-series constant. The former is a slope, the latter an intercept, and the present paper computes the latter.

2 The Castellated Indicator Function on $\mathbb{T}^1$

In the classical circle method [1], the distribution of integers is analyzed on the unit circle. Here, we map the local congruence condition modulo q ($q > 3$, prime) to a continuous phase variable $\theta \in [0, 1)$.

Definition 1 (Castellated Indicator Function). *For a given prime q and a constellation A , the forbidden residues (the ‘dead(q)’ mechanism) dictate topological exclusions. We define the Castellated Indicator Function $\Psi_{A,q}(\theta)$ as:*

$$\Psi_{A,q}(\theta) = \begin{cases} 0, & \text{if } \theta \in \bigcup_{r \in \text{dead}_A(q)} [\frac{r}{q}, \frac{r+1}{q}) \\ 1, & \text{otherwise} \end{cases} \quad (1)$$

For a Twin prime center ($6N \pm 1$), the forbidden residues are explicitly governed by $6N \equiv \pm 1 \pmod{q}$. This creates exactly two topological “holes” in the phase space. The base survival measure (single-center expectation) is simply the integral over the torus:

$$\mu_q = \int_0^1 \Psi_{\text{Twin},q}(\theta) d\theta = \frac{q-2}{q} \quad (2)$$

the complement of the microscopic root $P(q \mid N, \text{twin}) = 1/(q-2)$ of Part XII [3], re-expressed as a geometric transmitting measure.

3 The Circular Convolution Integral

We now consider the spatial entanglement of two Twin prime centers separated by distance j . The translation from center N to $N + j$ induces a precise phase shift in the q -frequency space:

$$\Delta\theta = \frac{j}{q} \quad (3)$$

The joint survival measure of both centers is equivalent to the overlapping “transmitting” area of their respective indicator functions. This is rigorously expressed as a continuous circular convolution integral on $\mathbb{T}^1$:

$$I_q(j) = (\Psi_q \circledast \Psi_q) \left(\frac{j}{q} \right) = \int_0^1 \Psi_q(\theta) \cdot \Psi_q \left(\left(\theta + \frac{j}{q} \right) \bmod 1 \right) d\theta \quad (4)$$

The correlation ratio $R_q(j)$, normalized against the independent probabilities, is therefore:

$$R_q(j) = \frac{I_q(j)}{\mu_q^2} = \frac{q^2}{(q-2)^2} \int_0^1 \Psi_q(\theta) \cdot \Psi_q \left(\theta + \frac{j}{q} \right) d\theta \quad (5)$$

Two clarifications. First, the integral $\int_0^1 \Psi_q(\theta) \Psi_q(\theta + s) d\theta$ is strictly the *autocorrelation* of Ψ_q ; it coincides with the circular convolution because the twin’s hole set $\{6^{-1}, -6^{-1}\}$ is centrally symmetric, so Ψ_q is even. Second, because the lag j is an integer the phase shift j/q is always a multiple of the cell width $1/q$, so this continuous integral is *exactly* the discrete count of co-surviving residues: the phase-space picture is a rigorous geometric reformulation of the singular-series ratio, not an approximation of it.

The global spatial correlation across the $6N$ lattice is the infinite product of these single-frequency phase-space integrals. Due to the rapid spectral decay of the interference amplitude, this product can be truncated at a spectral cutoff $k^*(\varepsilon)$ for any desired precision ε :

$$R(j) = \prod_{q>3}^{k^*} R_q(j) \quad (6)$$

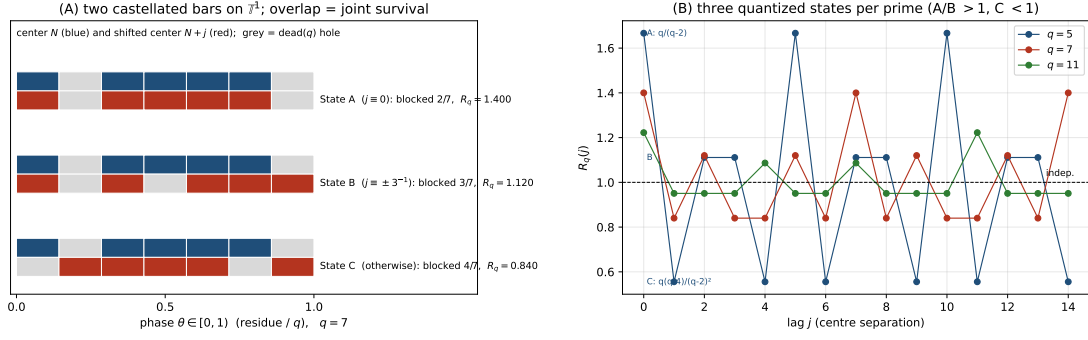


Figure 1: Left: the three interference states of Twin–Twin entanglement on $\mathbb{T}^1$ for $q = 7$. The castellated indicator of center N (blue) and of the shifted center $N + j$ (red) overlap; grey cells are the dead(q) holes. State A ($j \equiv 0$) blocks $2/q$, State B ($j \equiv \pm 3^{-1}$) blocks $3/q$, State C (otherwise) blocks the maximal $4/q$. Right: the resulting quantization of $R_q(j)$ for $q = 5, 7, 11$ into three discrete levels (A, $B > 1$; $C < 1$). At $j = 1$ every prime is in State C, so $R(1) = \prod_q R_q(1) = \mathfrak{S}_{\text{quad}} / \mathfrak{S}_{\text{twin}}^2 \approx 0.40$.

4 Quantization of Interference States

Because $\Psi_q(\theta)$ is a step function with two holes of width $1/q$, the continuous integral collapses into three discrete, quantized states, governed entirely by the geometric alignment of the holes.

The intrinsic phase difference between the two holes of a Twin prime is $\Delta_{\text{holes}} \equiv \pm 2 \cdot 6^{-1} \equiv \pm 3^{-1} \pmod{q}$.

Theorem 1 (The Three Interference States). *The circular convolution integral $R_q(j)$ evaluates to exactly three states:*

1. **State A (Constructive Interference):** When $j \equiv 0 \pmod{q}$. The phase shift is 0. The holes perfectly overlap. Total blocked area is $2/q$.

$$R_q(j) = \frac{(q-2)/q}{((q-2)/q)^2} = \frac{\mathbf{q}}{\mathbf{q}-2} > 1$$

2. **State B (Partial Interference):** When $j \equiv \pm 3^{-1} \pmod{q}$. The phase shift aligns one hole of the first center with the other hole of the second. Total blocked area is $3/q$.

$$R_q(j) = \frac{(q-3)/q}{((q-2)/q)^2} = \frac{\mathbf{q}(\mathbf{q}-3)}{(\mathbf{q}-2)^2} > 1$$

3. **State C (Destructive Interference):** For all other j . The holes are totally misaligned. Total blocked area is maximal at $4/q$.

$$R_q(j) = \frac{(q-4)/q}{((q-2)/q)^2} = \frac{\mathbf{q}(\mathbf{q}-4)}{(\mathbf{q}-2)^2} < 1$$

Remark on Spectral Decay: Observe that across all three states, as the spatial frequency $q \rightarrow \infty$, the correlation ratio asymptotically approaches unity: $\lim_{q \rightarrow \infty} R_q(j) = 1$. This rigorously proves that high-frequency topological holes become vanishingly small, and the spatial interference is overwhelmingly dominated by the lowest frequencies (small primes). This analytic property justifies the existence of the spectral cutoff $k^*(\varepsilon)$ defined in Equation (6) and guarantees the absolute convergence of the global lattice correlation product.

Quantitatively, the per-prime deviation from unity is $|\ln R_q| \approx 2/q$ for State A, $\approx 1/q$ for State B, and $\approx 4/(q-2)^2 = O(1/q^2)$ for State C. For any fixed lag j , every prime $q > 3j+1$ is in State C (only the finitely many primes dividing j or $3j \pm 1$ can be A or B), so the truncation tail is governed by the convergent sum $\sum_{q>k^*} 4/(q-2)^2$. Evaluating the product numerically for the canonical case $j=1$ (all State C) gives the cutoff $k^*(\varepsilon)$ at which the truncated product is within relative error ε of the full product $R(1) = 0.3969$:

ε	k^*	primes retained
10^{-1}	13	4
10^{-2}	73	19
10^{-3}	557	100
10^{-4}	4229	577

Table 1: Spectral cutoff $k^*(\varepsilon)$ for the truncated lattice correlation product at $j=1$. To 1% precision only the primes up to 73 are needed; the $O(1/q^2)$ State-C tail makes the product converge rapidly.

5 Limit Case Proof: The Quadruplet Collapse

To demonstrate the analytic power of this geometric formulation, we evaluate the limit case $j=1$, which physically corresponds to adjacent Twin centers merging into a Prime Quadruplet $(6N-1, 6N+1, 6N+5, 6N+7)$.

Let us test the spatial frequency $q=5$. We compute the phase difference: $3^{-1} \pmod{5} \equiv 2$. Since $j=1$, we have $1 \not\equiv 0 \pmod{5}$ and $1 \not\equiv \pm 2 \pmod{5}$.

Thus, $j=1$ deterministically triggers **State C (Destructive Interference)**. Substituting into the State C equation yields:

$$R_5(1) = \frac{5(5-4)}{(5-2)^2} = \frac{5}{9} \approx 0.555 \quad (7)$$

This is the largest single-prime suppression, and the product over all primes gives $R(1) = \prod_{q>3} R_q(1) = \mathfrak{S}_{\text{quad}}/\mathfrak{S}_{\text{twin}}^2 \approx 0.40$: the quadruplet is about $0.40\times$ as common as two independent twins. We are careful about what this proves. It is the suppression of the quadruplet *prefactor*—its X -independent singular series relative to the product of its constituents—and it is a deterministic consequence of four non-overlapping phase-space holes maximizing destructive interference on the integration manifold.

It is *not*, however, the -4 proper-depth slope of Volume I. The geodynamic law $\ln \rho_4 = \ln \mathfrak{S}_4 - 4\ell$ separates cleanly into an intercept $\ln \mathfrak{S}_4$ and a slope -4 ; the destructive interference computed here fixes the intercept (through $\mathfrak{S}_4$), while the slope -4 is the bare member count, the four factors of $1/\ln X$, independent of any inter-twin correlation. The two mechanisms are complementary and live on the orthogonal axes of §1: interference explains *why the constant is small*, dimension explains *why the density decays*. Conflating them would re-merge the very axes the framework is built to separate.

6 Conclusion

By translating modular arithmetic into the circular convolution of castellated indicator functions, we have established a pure continuous-integral foundation for spatial entanglement on the $6N$ lattice—the analytic form of the quasi-periodic congruence diffraction pattern identified empirically in the Volume I Review [6]. Future work will extend this 1D phase-space integration into multidimensional tensor products to compute the absolute spectral cutoff of the congruence diffraction pattern.

Data and code availability. The figure-generating script and an independent verifier (the three-state formula against direct union counts, and the $k^*(\varepsilon)$ error table) are openly available at <https://github.com/Ruqing1963/6N-phase-space> [7].

References

- [1] G. H. Hardy, J. E. Littlewood, *Some problems of ‘Partitio Numerorum’; III*, Acta Math. **44** (1923), 1–70.
- [2] R. Chen, *Prime distribution on the $6N$ skeleton (Part I)*, Zenodo, doi:10.5281/zenodo.20470367 (2026).
- [3] R. Chen, *The microscopic root $1/(q - 2)$ and the mean shift (Part XII)*, Zenodo, doi:10.5281/zenodo.20528446 (2026).
- [4] R. Chen, *Arithmetic geodynamics on the $6N$ skeleton (Part XVII)*, Zenodo, doi:10.5281/zenodo.20541575 (2026).
- [5] R. Chen, *The tuple-size law at $m = 5$: resolving the sparse regime (Part XIX)*, Zenodo, doi:10.5281/zenodo.20551393 (2026).
- [6] R. Chen, *Arithmetic geodynamics on the $6N$ skeleton: a review of Parts I–XIX*, <https://github.com/Ruqing1963/6N-review> (2026).
- [7] R. Chen, *The circular convolution of prime constellations in congruence phase space: code (Part XX)*, <https://github.com/Ruqing1963/6N-phase-space> (2026).